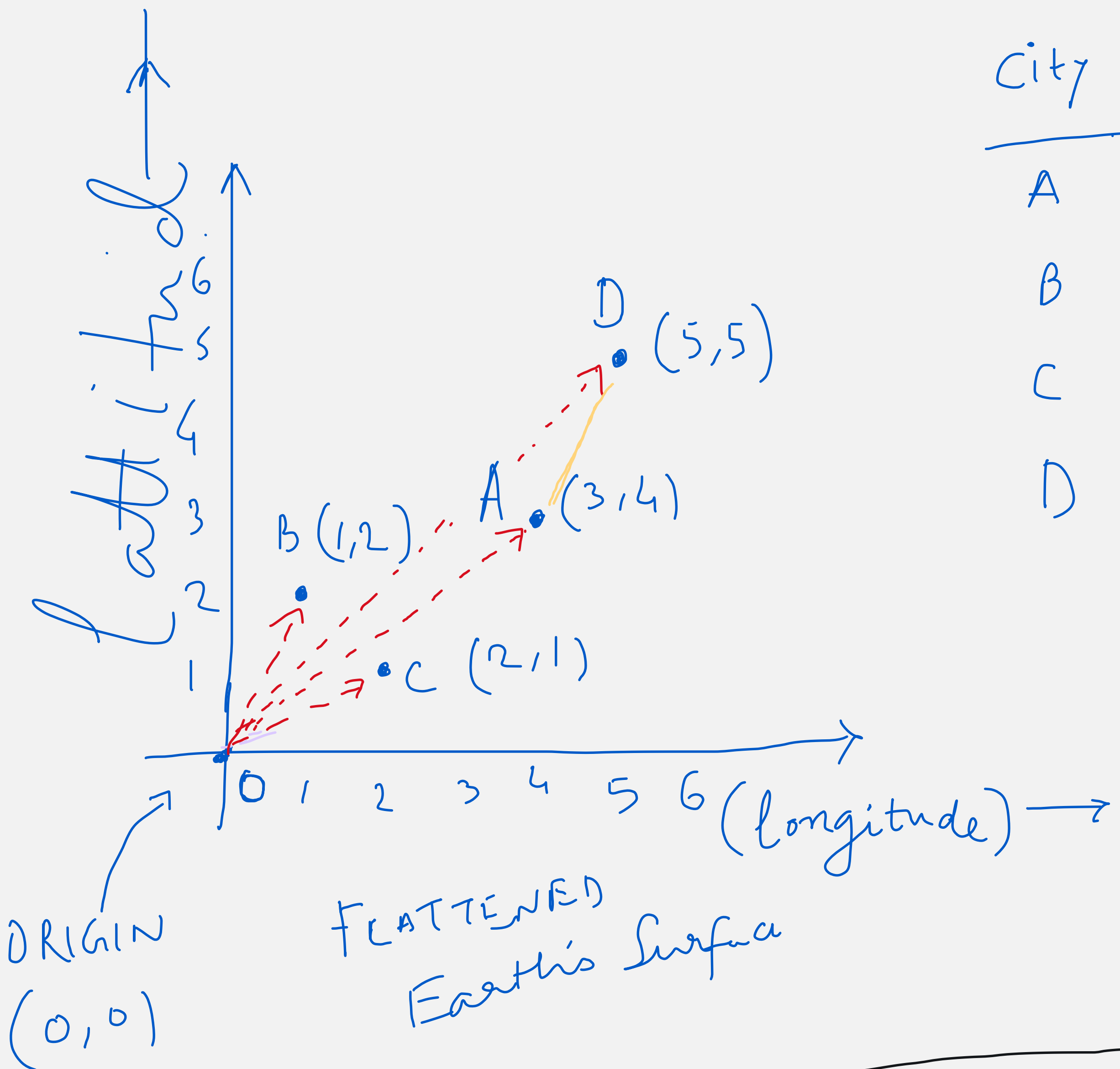

City	Longi.	Latti	vector
A	3	4	(3,4)
B	1	2	(1,2)
C	2	1	(2,1)
D	5	5	(5,5)

Norm(A) = distance of A from the origin

$$= \sqrt{(3-0)^2 + (4-0)^2}$$

$$= \sqrt{25} = 5$$

$$\text{dot product}(A,C) = x_1 \cdot x_2 + y_1 \cdot y_2$$

$$= 3 \times 2 + 4 \times 1$$

$$= 10$$

$$\text{dist}(A,C) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

$$= \sqrt{(3-2)^2 + (4-1)^2}$$

$$= \sqrt{1+9}$$

$$= \sqrt{10} = 3.16$$

$$\text{norm}(C) = \sqrt{2^2 + 1^2} = \sqrt{5} = 2.23$$

$$\text{Cosine}(A,C) = \frac{\text{dot}(A,C)}{(\text{norm}(A) \times \text{norm}(B))}$$

$$= \frac{10}{5 \times 2.23}$$

$$= 0.896$$